Same checked facts. Different answer.

Gemma-2-9B-it reads two starting versions of the same stance sentences. Both receive the same three update requests. Both then answer the fact checks correctly. Yet they answer one combined question differently.

“Do Ada and Bryn take the same side on Gallery?”

History H5

No correct

History H6

Yes incorrect

DEFINITION **History:** one starting version of the stance sentences supplied to the model.

H5 and H6 are two of eight histories. One measured example, not an estimate of how often this happens.

The whole experiment in four steps

1 DIFFERENT HISTORIES

Eight starting versions of the same stance sentences.

2 ONE SET OF REQUESTED FINAL FACTS

Every history receives the same three update requests.

3 INDEPENDENT QUESTIONS

Each question is asked on its own copy. None sees another’s answer.

4 COMPARE CHECKED ANSWERS

Do the answers depend only on the final facts, or also on the starting history?

DEFINITION

Matched histories: different starting histories given the same requested final facts, then compared using the same update procedure and questions.

The intended final facts

Person	Garden proposal	Meadow proposal	Gallery proposal
Bryn	− Opposed	+ In favor	− Opposed
Ari	+ In favor	+ In favor	− Opposed
Ada	− Opposed	+ In favor	+ In favor
Dara	− Opposed	− Opposed	− Opposed

Only the three outlined facts differ between starting histories; the other nine never change.

These are fictional facts stated in the text Gemma is given, not knowledge Gemma already had.

Prompt, part 1 of 2: two solved demonstrations

```
<bos><start_of_turn>user
Example 1:
This is a fictional community planning meeting. Use only
the records below.
Each named member has independent views. Updating one
record leaves the other records unchanged.
Ravi is in favor of the Gallery proposal.
Ravi is opposed to the Harbor proposal.
Pia is opposed to the Gallery proposal.
Pia is opposed to the Harbor proposal.
Esme is opposed to the Gallery proposal.
Esme is opposed to the Harbor proposal.
Bela is in favor of the Gallery proposal.
Bela is opposed to the Harbor proposal.
End of records.
Does Esme favor the Harbor proposal?
Answer with exactly one of No, Yes.
Answer: No
```

```
Example 2:
This is a fictional community planning meeting. Use only
the records below.
Each named member has independent views. Updating one
record leaves the other records unchanged.
Gita is opposed to the Garden proposal.
Gita is in favor of the Library proposal.
Juno is opposed to the Garden proposal.
Juno is in favor of the Library proposal.
Mina is opposed to the Garden proposal.
Mina is in favor of the Library proposal.
Esme is in favor of the Garden proposal.
Esme is in favor of the Library proposal.
End of records.
Does Esme favor the Library proposal?
Answer with exactly one of No, Yes.
Answer: Yes
```

Every prompt opens with these two demonstrations. They are the same in every history and for every question.

Prompt, part 2 of 2: the task input for history H5

```
Now the actual records:
This is a fictional community planning meeting. Use only the records
below.
Each named member has independent views. Updating one record leaves the
other records unchanged.
Bryn is in favor of the Gallery proposal. ← old
Bryn is opposed to the Garden proposal.
Bryn is in favor of the Meadow proposal.
Ari is in favor of the Garden proposal.
Ari is in favor of the Meadow proposal.
Ari is opposed to the Gallery proposal.
Ada is opposed to the Garden proposal.
Ada is in favor of the Meadow proposal.
Ada is opposed to the Gallery proposal. ← old
Dara is opposed to the Garden proposal.
Dara is opposed to the Meadow proposal.
Dara is opposed to the Gallery proposal.
End of records.
── a question is appended on a separate copy ──
Do Ada and Bryn take the same side on Gallery?
Answer with exactly one of No, Yes.
Reply with only that answer on the first line.<end_of_turn>
<start_of_turn>model
```

Twelve stance sentences. In H5, two still state the old Gallery stances.

Not in this text: any update request or correct answer.

455 tokens before the question, in H5. The question adds 37.

Teal line: annotation, not model text.

Eight starting histories, one set of final facts

STARTING STANCE ON GALLERY · all eight are evaluated

	Bryn	Ari	Ada
H0	− Opposed	− Opposed	+ In favor
H1	+ In favor	− Opposed	+ In favor
H2	− Opposed	+ In favor	+ In favor
H3	+ In favor	+ In favor	+ In favor
H4	− Opposed	− Opposed	− Opposed
H5	+ In favor	− Opposed	− Opposed
H6	− Opposed	+ In favor	− Opposed
H7	+ In favor	+ In favor	− Opposed

REQUESTED FINAL FACTS · the same for all eight

Bryn	− Opposed
Ari	− Opposed
Ada	+ In favor

Dark outline = differs from the requested final fact. H5 and H6 are the worked pair.

Every history receives the same three requests

THE THREE UPDATE REQUESTS · each about the Gallery proposal

	Bryn → − Opposed	Ari → − Opposed	Ada → + In favor	facts to change
H5	changes the fact	already correct; still runs	changes the fact	2
H6	already correct; still runs	changes the fact	changes the fact	2

All three requests run in every history, even in H0, where every fact is already correct. A request states the intended fact; it does not prove what changed inside the model.

What kind of update is used here?

ACTIVATION INTERVENTION

Used in this example

The sentences stay exactly as written. A trained function changes the activations of the addressed sentence.

APPENDED TEXT CORRECTION

A different update route

A correction sentence is added to the prompt. Gemma reads the old and the new text.

FINAL FACTS FROM THE START

A comparison run, not an update

The prompt states the final facts from the beginning. Nothing is updated.

The next slide shows the appended text correction on this same example.
Every later slide follows the activation intervention.

Appended text correction: a worked example

```
Now the actual records:
This is a fictional community planning meeting. Use only
the records below.
Each named member has independent views. Updating one
record leaves the other records unchanged.
Bryn is in favor of the Gallery proposal. ← old
Bryn is opposed to the Garden proposal.
Bryn is in favor of the Meadow proposal.
Ari is in favor of the Garden proposal.
Ari is in favor of the Meadow proposal.
Ari is opposed to the Gallery proposal.
Ada is opposed to the Garden proposal.
Ada is in favor of the Meadow proposal.
Ada is opposed to the Gallery proposal. ← old
Dara is opposed to the Garden proposal.
Dara is opposed to the Meadow proposal.
Dara is opposed to the Gallery proposal.
```

```
Correction: the record for Bryn on the Gallery proposal is
updated. Bryn is now opposed to the Gallery proposal.
Correction: the record for Ari on the Gallery proposal is
updated. Ari is now opposed to the Gallery proposal.
Correction: the record for Ada on the Gallery proposal is
updated. Ada is now in favor of the Gallery proposal.
End of records.
— a question is appended on a separate copy —
Do Ada and Bryn take the same side on Gallery?
Answer with exactly one of No, Yes.
Reply with only that answer on the first line.<end_of_turn>
<start_of_turn>model
```

Three correction sentences are appended. Ari's stance was already correct; its correction is still appended.

A demonstration of the textual update route on H5. The result in this deck was measured with the activation intervention, not with this prompt.

The two demonstrations from part 1 come first, unchanged. Blue: old statements that remain in the text.

From the sentence to its token positions

1 A KNOWN SENTENCE

“Bryn is in favor of the Gallery proposal.”

the addressed sentence, at a known place in the H5 prompt

2 TOKEN POSITIONS



The tokenizer maps this addressed sentence to 10 token positions.

3 ACTIVATIONS



one activation per token position, 3,584 numbers each

The code is told where the sentence is and writes there. It has not searched for where a fact is stored.

The three addressed sentences in H5 span 10, 8 and 8 token positions.

Fixed model weights, temporary activations

MODEL WEIGHTS · FIXED

Gemma’s parameters. Nothing in this experiment changes them.

ACTIVATIONS · TEMPORARY

Computed fresh for each prompt: 3,584 numbers per token. The intervention changes these.

Blocks 17–42

continue from the changed activations

After block 16

the intervention changes the addressed sentence

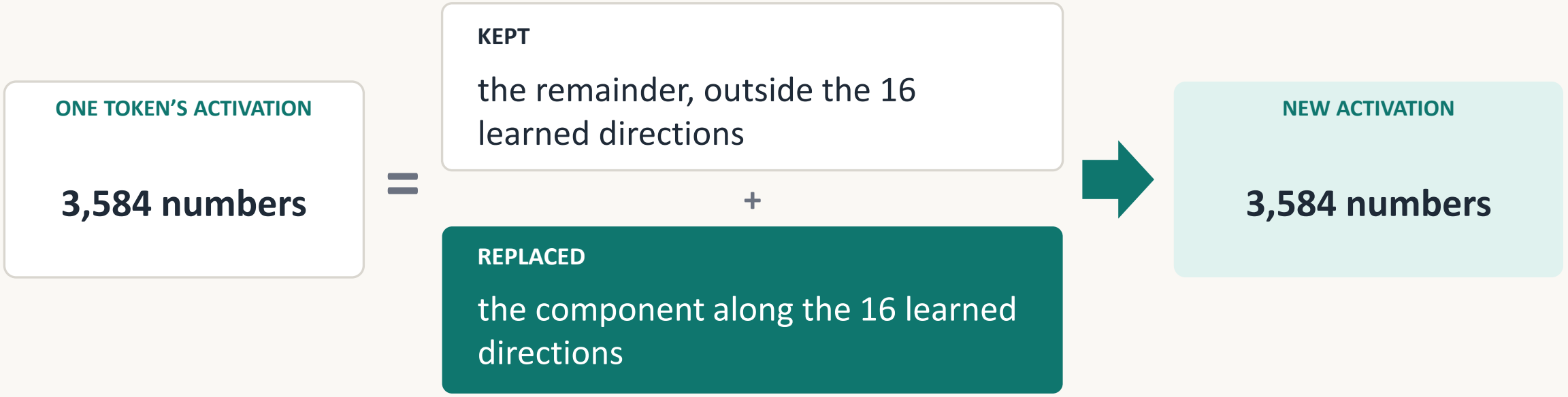
Blocks 1–15

run on the unchanged text

↑ the prompt enters at the bottom

Fixed setting: the output of block 16, at the addressed sentence only. Not at question or answer positions.

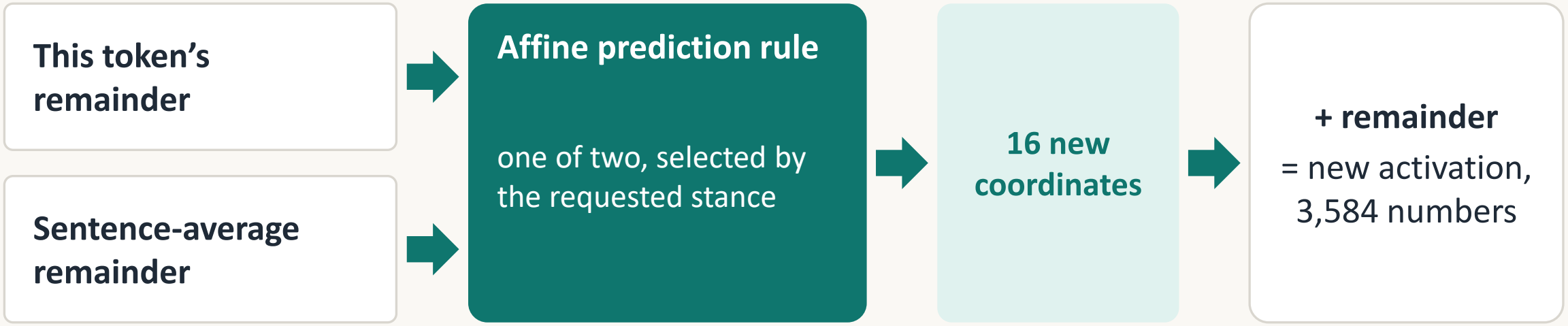
Low rank: a new component along 16 learned directions



Each learned direction is a pattern across all 3,584 numbers, not 16 of the numbers themselves.

Low rank: the intervention can change only the component along these 16 directions.

How the replacement is calculated



A coordinate is the amount along one learned direction. In H5, Bryn's 10 token activations go in and 10 replacements come out.

A small numerical function: one matrix multiplication and a bias. Not another language model.

Same saved rule and directions in every history. Different incoming activations can give different replacements.

How the intervention was trained, before this test

TEXT GEMMA READS · two starting versions

```
Mina is in favor of the Bridge proposal.
Mina is opposed to the Bridge proposal.
```

plus seven other stance sentences, the same in both

UPDATE REQUEST · given separately

Mina on Bridge → Opposed

an input to the intervention, not text Gemma reads

CORRECT ANSWER · training only

No

to “Does Mina favor the Bridge proposal?”

Intervention → **Gemma · weights frozen** → **a training question** → **probability of the correct answer**

gradients pass back through frozen Gemma and update only the intervention

Trained jointly: the 16 directions and both prediction rules. Training questions include combined ones.

64 training pairs → **32 separate pairs pick the saved version** → **frozen** → **fresh tests (this deck)**

Ask every question independently

1 PREPARE THE CONTEXT ONCE

Gemma reads the text with the three interventions applied. No question exists yet.

2 SAVE THE KV CACHE

one master cache for each history

- **fresh clone + question 1 → answer**
- **fresh clone + question 2 → answer**
- **fresh clone + question 3 → answer**

Each question gets a fresh clone of the same cache, and the clone is then discarded. No question or answer changes the master cache. A text update builds the context differently; questions are then served the same way.

6 direct “favor” + 2 direct “oppose” + 9 combined = 17 questions
17 questions × 2 answer formats = 34 scored questions per history

Answer formats: “No / Yes”, or letters set for each question, e.g. “Use A for yes and B for no.”

Every combined question gets a turn

	Same side?	Both in favor?	At least one in favor?
Ada & Bryn	**No**	**No**	**Yes**
Bryn & Ari	**Yes**	**No**	**No**
Ari & Ada	**No**	**No**	**Yes**

Cells show the correct answers about Gallery. Solid outline: the question being tested in our example. Dashed: the next turn.

Each combined question gets one turn as the question being tested.

For that turn, the other 8 combined questions become its checks.

Then the roles rotate. No new questions are asked: the same saved answers are reused.

9 combined questions × 2 answer formats = 18 scheduled questions

Checks come from the same 34 answers: the 12 direct “favor” responses and the other combined questions.

Why No is correct, and what Gemma answered

"Do Ada and Bryn take the same side on Gallery?"

Ada + In favor Bryn – Opposed

Different sides → correct answer: No

H5 answers

No ✓ correct

No		0.78
Yes		0.22

H6 answers

Yes ✗ incorrect

No		0.02
Yes		0.98

Bars: probability that the next token is "No" or "Yes". Not a measure of how sure the model is.

H0	H1	H2	H3	H4	H5	H6	H7
No ✓	No ✓	Yes ✗	No ✓	Yes ✗	No ✓	Yes ✗	No ✓

What the H5/H6 comparison holds equal

	H5	H6
Facts the requests must change	2	2
Tokens before the question	455	455
Update requests	the same three	
Question	the same	

WHY MATCH?

This rules out two simple explanations: one history needing more changes, or one history being longer.

Still different: which starting stances belonged to whom.

Direct fact checks and two comparison runs

DO THE UPDATED FACTS READ BACK? · six facts asked directly

Does Bryn favor the Gallery proposal?	No ✓
Does Ari favor the Gallery proposal?	No ✓
Does Ada favor the Gallery proposal?	Yes ✓
Does Dara favor the Gallery proposal?	No ✓
Does Bryn favor the Garden proposal?	No ✓
Does Ada favor the Garden proposal?	No ✓

Two answer formats each: 12 responses, correct in all 8 histories.

TWO COMPARISON RUNS · the same question

1 CAN GEMMA ANSWER IT?

Final facts from the start →
no intervention → same question

No ✓

2 DOES THE UPDATE ITSELF BREAK IT?

Final facts from the start (H0) →
same 3 interventions → same question

No ✓

Both comparison runs pass. So the H5/H6 disagreement is not explained by an unanswerable question, or by an update that always breaks the answer.

The other combined questions also pass

	Same side?	Both in favor?	At least one in favor?
Ada & Bryn	being tested	No ✓	Yes ✓
Bryn & Ari	Yes ✓	No ✓	No ✓
Ari & Ada	No ✓	No ✓	Yes ✓

Can Gemma combine these same facts? All eight are correct in all eight histories.

FOR THIS ONE TESTED QUESTION

12 direct responses
+ 8 other combined responses
= 20 checks per history
× 8 histories

= 160 **correct check responses**

(the two comparison runs are separate)

The next tested question gets its own 20 checks, reusing the same saved answers.

Not both in favor, but at least one in favor ⇒ different sides.
Yet asked separately, H6 answers “Same side?” with Yes.

These answers conflict with H6’s answer to the question being tested.

How the result is counted

THE COUNTING RULE · fresh study of 64 new cases

A scheduled question counts once when its checks and comparison runs pass and a matched pair of histories gives different answers.

All 18 scheduled questions stay in the denominator, including those whose checks fail.

2 of 18	in this example
0.95%	of scheduled questions, across all 64 cases
14 of 64	cases with at least one

The broader figures combine two separately trained versions of this intervention. They describe this controlled test, not real-world frequency and not a ranking of methods.

What the matched-histories test establishes

OBSERVATION

H5 and H6 receive the same requested final facts and the same update procedure.
Both pass the required fact and combined-question checks.
They give different answers to the same combined question.

WHAT THIS SUPPORTS

Passing those checks is not sufficient to establish that later answers are independent of the starting history. Under this update procedure, the later answer can still vary with it.

WHY MATCHED HISTORIES MATTER

A generic later error need not differ between histories. This test asks whether the starting history predicts a difference once the intended facts are aligned and checked.

Scope: this is a behavioral result.
It does not identify the internal mechanism that causes the difference.